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Effect of Hf on high-temperature strength and room-temperature ductility of Nb–15W–0.5Si–2B alloy

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Abstract

Mechanical responses of Nb–15W–0.5Si–2B–*x*Hf alloys ($x = 0$, 5, 10 and 15 mol%) were investigated to probe the balance between room-temperature ductility and high-temperature strength depending on Hf concentration in Nb-based solid solution. It is found that Hf addition considerably improves the high-temperature strength of the Nb–15W–0.5Si–2B alloy without loss of the room-temperature ductility. With a Nb solid-solution dominant microstructure, the Nb–15W–0.5Si–2B–15Hf alloy shows a 0.2% compressive yield strength of as high as 365 MPa even at 1400 °C, while its compressive ductility at room temperature is about 17%.

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Keywords: Niobium alloys; Compression test; High-temperature strength; Room-temperature ductility

1. Introduction

Niobium (Nb) is one of the most promising refractory metals to be used as a base metal for high-temperature structural materials due to its relatively high melting point of 2470 °C and good room-temperature ductility. In order to promote high-temperature strength of the Nb-based alloys, a dual-phase Nb_{SS}/Nb_5Si_3 (ductile/brittle) structure [1–8] was employed as a concept for the microstructure design of the Nb-based alloys. With excellent high-temperature strength, however, the Nb_{SS}/Nb_5Si_3 structure with a silicide matrix microstructure (corresponding to Nb–18Si-based alloy) showed insufficient room-temperature ductility, as low as 1%. It is, thus, expected that the single metallic phase Nb_{SS} can obtain a relatively higher strength at the elevated temperatures by strong solid-solution hardening without considerable loss of the room-temperature ductility. In an earlier work [6], we prepared a Nb–10Mo–10Ti–15W–18Si in-situ composite with a yield strength of 350 MPa at 1500 °C. To further enhance the potential of the solid-solution hardening and the room-temperature ductility, Si content was reduced to 0.5 mol% to avoid formation of the Nb_5Si_3. 15 mol% W was maintained because W was most efficient for improving the high-temperature strength of the Nb_{SS} phase [6,7]. Hf with a melting point of 2754 K and a larger atomic size misfit of as high as 9% to Nb was also greatly expected to harden the Nb_{SS}. Two percent B was added also for the solid-solution hardening. Then a Nb–15W–0.5Si–2B–*x*Hf system is developed to probe the role of the Hf concentrations on the room-temperature ductility and the high-temperature strength of the Nb_{SS}-dominant microstructure.

2. Experimental procedures

Nominal composition of Nb–15W–0.5Si–2B mol% was used as the base and 0, 5, 10, and 15 mol% hafnium were respectively added. Elements with 99.9 mol% or higher purity were used. Button ingots were prepared by arc-melting five or six times in vacuum. The ingots were finally annealed at 1700 °C for 50 h. The microstructure was observed using scanning electron microscopy (SEM). Compressive tests were conducted at room temperature, 1200 °C and 1400 °C in vacuum at a strain rate of $3 \times 10^{-4}\ s^{-1}$ using an Instron 8560 testing machine. The dimensions of the polished compressive cylinders were 3 mm in diameter and 5 mm in length.

3. Results and discussions

3.1. Microstructure

Fig. 1 shows that the microstructures of Nb–15W–0.5Si–2B and Nb–15W–0.5Si–2B–15Hf alloys exhibit no visible

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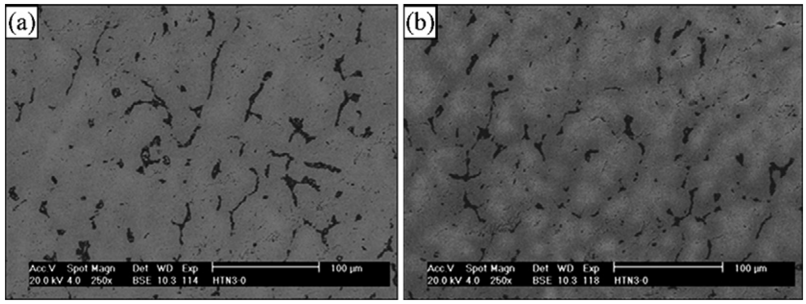

Fig. 1. Microstructures of Nb: Nb–15W–0.5Si–2B (a) and Nb–15W–0.5Si–2B–15 Hf (b).

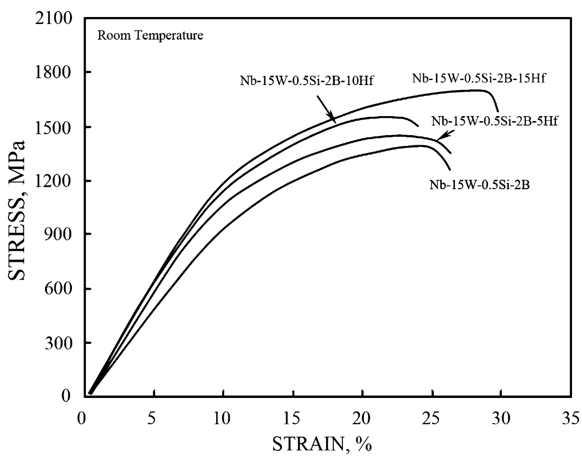

Fig. 2. Compressive stress–strain curves of Nb–15W–0.5Si–2B–(0–15)Hf alloys at room temperature.

difference when Hf concentration is changed from 0% to 15%. The microstructure of Nb–15W–0.5Si–2B–(0–15)Hf alloy is composed of two phases. The overwhelming phase is the Nb solid-solution, Nb_{SS}, and the minority grey phase is the Nb–B boride with a volume fraction less than 5%. No Nb_5Si_3 and Hf-based solid solution were observed. Hf, W and Si solidified entirely in the Nb_{SS}.

3.2. Compressive stress–strain curve at room temperature

The engineering stress–strain curves of the Nb–15W–0.5Si–2B–(0–15)Hf alloys at room temperature are shown in Fig. 2. Table 1 summarizes the 0.2% yield strength, $\sigma_{0.2}$, the maximum strength, σ_{max}, and the compressive ductility, ε_{max}, as a function of the Hf concentration, where ε_{max} is defined as the plastic strain corresponding to σ_{max}. It seems that $\sigma_{0.2}$ is independent of Hf below 10 mol% Hf; however, it exhibits about 10% increment for an Hf content of 15 mol%. σ_{max} increases monotonously exhibiting a 20% increase at

Table 1
Mechanical properties of the Nb–15W–0.5Si–2B–(0–15)Hf alloys as a function of the Hf contents at room temperature

Alloy	0Hf	5Hf	10Hf	15Hf
$\sigma_{0.2}$ (MPa)	775	780	793	870
σ_{max} (MPa)	1365	1430	1525	1680
ε_{max} (%)	12.0	13.0	11.5	17.0

Table 2
Mechanical properties of the Nb–15W–0.5Si–2B–(0–15)Hf alloys at high temperatures

	Alloy			
	0Hf	5Hf	10Hf	15Hf
1400 °C				
$\sigma_{0.2}$ (MPa)	180	265	360	365
σ_{max} (MPa)	–	–	–	–
1200 °C				
$\sigma_{0.2}$ (MPa)	295	375	410	448
σ_{max} (MPa)	590 (at a strain of 28%)	780 (at a strain of 28%)	775	925

15 mol% Hf. It should also be emphasized that, the ε_{max} of the Nb–15W–0.5Si–2B–(0–15)Hf alloys have not been degraded in spite of the solid-solution hardening of Nb_{SS} by Hf addition. The ε_{max} of the Nb–15W–0.5Si–2B–(0–15)Hf alloys are in a range of 11.5–17%, which is valuable for the engineering applications compared with those of the Nb-based alloys with a Nb_5Si_3 matrix or network structure [6,7].

3.3. Mechanical behavior at elevated temperatures

The compressive stress–strain curves of the Nb–15W–0.5Si–2B–(0–15)Hf alloys at high temperatures are shown in Fig. 3. At 1400 °C (see Fig. 3(a)), four samples show a strong deformation hardening up to a plastic strain of 5%, followed by a low and linear work hardening rate until the plastic strain is larger than 20%. With increasing Hf content, the stress clearly exhibits a higher level at the same strain. Generally, the stress–strain curves of the four samples deformed at 1200 °C exhibit the same tendency as those deformed at 1400 °C, as seen in Fig. 3(b). It is known from Table 2 that at 1400 °C the $\sigma_{0.2}$ is dramatically increased by a 15 mol% Hf addition from 180 MPa for the Hf-free sample, to 365 MPa. At 1200 °C, $\sigma_{0.2}$ is increased by 50% for a Hf addition of 15 mol%.

4. Discussion

Fig. 1 shows that the morphology and the volume fraction of the boride are independent of the Hf content. This suggests that the effect of the boride on the bulk properties of each alloy is at the same level. Hence, the bulk property changes of these alloys can be attributed to the Hf addition in the Nb_{SS} when the

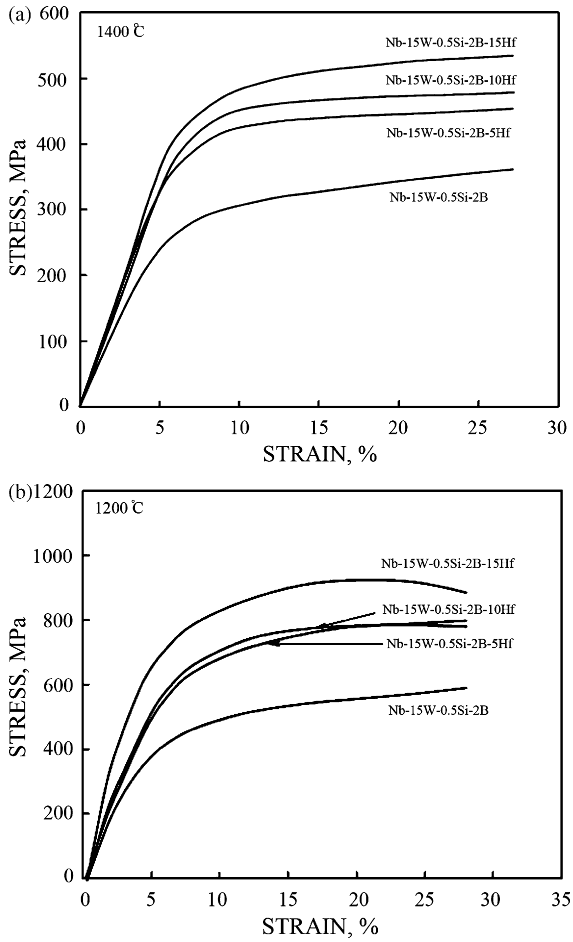

Fig. 3. Compressive stress–strain curves of Nb–15W–0.5Si–2B–(0–15)Hf alloys at elevated temperatures: (a) 1400 °C and (b) 1200 °C.

nominal contents of W, Si and B are fixed. From Tables 1 and 2, it is known that the Hf addition benefits to both the strength and ductility of the Nb_{SS}. Three factors affect the solid-solution hardening, namely, the structural difference between Nb and the alloying element, the size misfit and the solubility limit [9] of the alloying element in Nb. The larger the size misfit, the higher the solid-solution-hardening rate is. In the current Nb–W–Si–B–Hf system, W and Si have a positive effect on the solid-solution hardening of the Nb_{SS} [6,7]. The solid-solution hardening on Nb supplied by Hf is mainly attributed to the large size misfit of about 9%, and to the structural difference between bcc (Nb) and hcp (Hf). With such a large size misfit, the alloy lattice parameter is increased from 0.3327 nm for the Nb–15W–0.5Si–2B alloy, to 0.3407 nm for 15 mol% Hf, and the Vickers hardness, HV, is increased linearly from 294 for the Hf-free sample, to 400 for the alloy with a 15 mol% Hf content. Therefore, with a 15 mol% Hf addition, the bcc Nb_{SS} obtains a 2.4% increment in the lattice parameter, which may achieve a strong elastic interaction for non-screw dislocations.

The present Nb–15W–0.5Si–2B–(10–15)Hf alloys have about the same strength level as the Nb–10W–10Si alloy containing about 27% Nb_5Si_3 (i.e. a 0.2% yield strength of 330 MPa at 1670 K [7]). The former, however, has room-temperature ductility higher than 11%, while the room-temperature ductility of the latter is negligibly small. Hence, Nb_5Si_3 free Nb-based alloys already have enough strength to replace the Nb_{SS}/Nb_5Si_3 structure for high-temperature applications, even with less than 5% Nb–B boride.

In this work, Hf is beneficial to both the strength and ductility of the Nb_{SS} at variance from the relationship between the strength and ductility of general engineering materials. Chan [10] pointed out that Hf and Ti are the only alloying elements that do not increase the BDTT of the Nb_{SS}, and on the other hand, that Hf and Ti are the only toughening and ductilizing elements for Nb. The effect of Ti addition on the toughening and ductilization of the Nb_{SS} has been modeled [11] by considering unstable stacking energy and the Peierls–Nabarro barrier energy. The theoretical calculations indicate that the unstable stacking energy in Nb_{SS} is insensitive to Ti addition. In contrast, the Peierls–Nabarro barrier energy for the emission of edge dislocations on (1 1 0) and (1 1 2) decreases with increasing Ti addition. The lowering of the Peierls–Nabarro barrier energy by Ti addition increases both the fracture toughness and ductility of the Nb_{SS}.

Because Hf and Ti have the same crystal structure (hcp) and both of them show the same propensity towards ductilization of the Nb_{SS}, the ductilization of the Nb_{SS} by Hf may also be dominated by either one of the unstable stacking energy and the Peierls–Nabarro barrier energy or both. The corrective explanation of the ductilization of the Nb_{SS} by Hf, however, could be given only when the theoretical calculations combine with the TEM observations.

5. Conclusions

In niobium alloyed with W, Si, B and Hf, the following conclusions were reached:

(i) The Nb–15W–0.5Si–2B–(0–15)Hf alloys display a Nb solid-solution dominant microstructure.
(ii) An outstanding solid-solution hardening on the Nb–15W–0.5Si–2B alloy at room and high temperatures is achieved by Hf addition, without degradation of the room-temperature ductility. At 1400 °C, the yield stress increases from 180 MPa, for the Hf-free alloy, to 365 MPa, for the alloy containing 15 mol% Hf. The compressive ductilities of the above two alloys at room temperature are about 12% and 17%, respectively.

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